BANQUET HALL BOOKING SYSTEM

Smart Venue Reservation & Event Management Platform

Phase 4 — Project Design

Problem-Solution Fit | Proposed Solution | Architecture

4.1 Problem-Solution Fit

The following matrix maps each identified problem to its corresponding solution feature in the Banquet Hall Booking System:

Problem	Solution Feature
Clients cannot find and compare venues efficiently without in-person visits.	Centralized search portal with advanced filters, photos, 360-degree virtual tours, and real-time availability.
Double bookings and scheduling conflicts due to manual management.	Automated booking calendar with conflict detection and real-time synchronization across all devices.
Opaque pricing and hidden costs create distrust between clients and venues.	Transparent pricing engine with itemized package breakdowns and instant quotes.
Communication gaps delay event planning and cause last-minute issues.	In-app messaging, automated notifications (email/SMS), and shared event coordination workspace.

4.2 Proposed Solution

The proposed Banquet Hall Booking System is a multi-tenant SaaS platform with three distinct user roles: Client, Venue Manager, and System Administrator. The solution encompasses the following core modules:

Venue Discovery Module	Booking Engine	Payment Gateway

Search, filter, and shortlist halls based on personalized criteria — date, capacity, location, amenities, and price.	Step-by-step guided booking wizard with package selection, catering options, and add-on services.	Secure multi-mode payments with advance deposit, balance tracking, and auto-generated invoices.
Event Coordination	**Analytics Dashboard**	**Review System**
Shared event planning workspace for clients and managers to finalize decor, catering, and logistics.	Revenue analytics, occupancy rate, booking trends, and performance insights for venue managers.	Verified post-event reviews and ratings to maintain quality and build trust in the marketplace.

4.3 Solution Architecture

The system follows a three-tier architecture pattern with clear separation of concerns between presentation, business logic, and data layers.

Presentation Layer React / Next.js	→	**API Gateway REST / GraphQL**	→	**Business Logic Node.js / Django**	→	**Data Layer MongoDB / PostgreSQL**

Architecture Pattern: MVC (Model-View-Controller) with microservices for payment processing, notifications, and search.

Scope Definition

✅ In Scope	❌ Out of Scope
• Online hall search and discovery • Real-time booking calendar • Secure payment integration • Email/SMS notifications • Venue manager dashboard • Client profile and booking history • Post-event reviews	• On-site catering kitchen management • Third-party vendor management • Multi-location event logistics • Physical equipment inventory • Live video streaming of events • HR or staff payroll management